N83 K128 split root index: run record

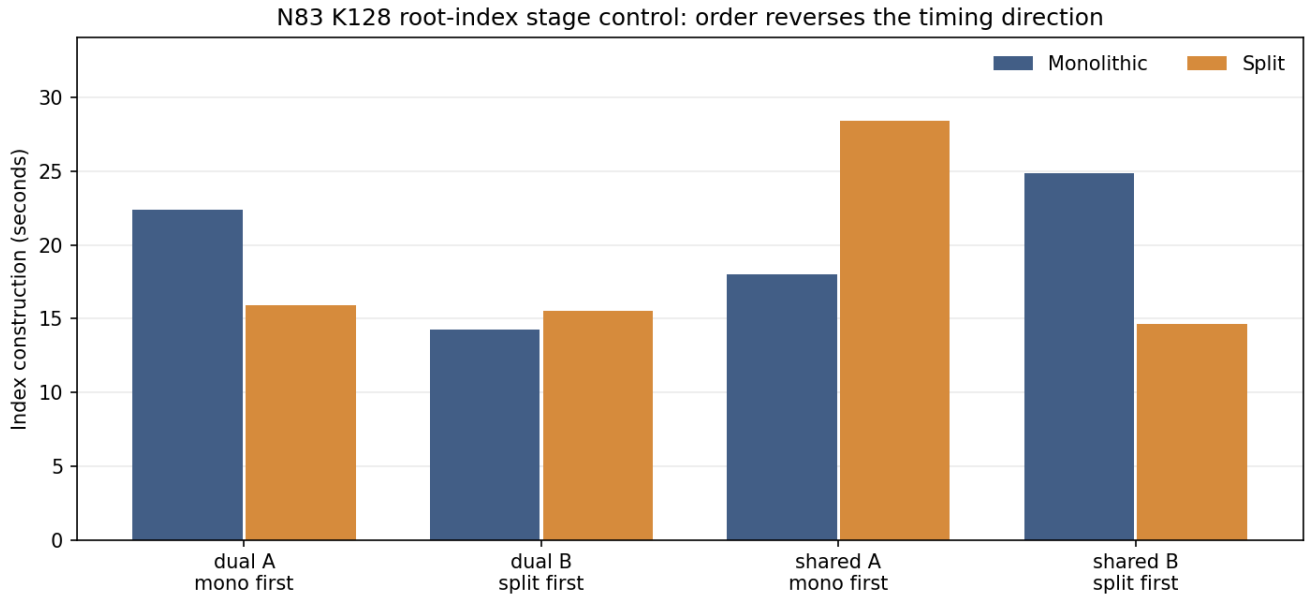
Four paired index-stage controls built the same 21,248-point N83 factor base and 680,000 symmetric states. Each pair used one executable in alternate monolithic/split order. The public target file was fixed but target work was never started in these stage controls.

All four process receipts passed. Each arm inserted 1,359,488 exact root entries and retained the same base hash and state digest. The split point was K64. The dual variant used per-table Bloom filters; the shared variant used one union filter. Local shared macOS timing is a stage diagnostic.

Source revision and executable hashes are frozen in each input.json. Raw results are in the study directory, and independent exact-root unit controls passed in the example test suite (22 tests).

A further K128 control set KIC_INDEX_ROOT_THREADS=4. Its two layouts matched the serial base, state, exact-table, root-entry, allocation, and Bloom fields under independent replay. The four-thread release binary and prior serial debug binary are used for output identity only.

Order	Mode	Mono (s)	Split (s)	Faster stage
k128_a	dual	22.393	15.919	split
k128_b	dual	14.285	15.502	mono
k128_shared_a	shared	18.030	28.390	mono
k128_shared_b	shared	24.862	14.632	split



Source: research/n83_two_tier_root_20261021/runs/{k128_a,k128_b,k128_shared_a,k128_shared_b}/result.json; same study directory contains inputs, stderr, stdout, and runner copies.